

Title

Impact of cholera outbreak response immunization: Insights from modeling outbreaks in sub-Saharan Africa, 2010-2020.

Abstract

Background

Cholera outbreaks remain a major public health threat in sub-Saharan Africa. The impact of oral cholera vaccines (OCVs) depends on deployment conditions. We evaluated how the health and economic impact of single-dose outbreak response immunization (ORIs) varies across outbreak and implementation parameters.

Methods

Using data from 1,192 cholera outbreaks (2010–2020) in sub-Saharan Africa, we simulated counterfactual ORIs varying by timing, coverage, triggering criteria, and target outbreak selection. Outcomes—evaluated at the individual-outbreak level and aggregated across subsets—included *impact* (cases, deaths, and DALYs averted), *efficiency* (impact per 1,000 OCV doses), and *cost-effectiveness* (cost per unit of impact). Stochastic simulations incorporated age-specific vaccine efficacy, indirect protection, delays in ORI initiation, and the time required to develop vaccine-derived immunity.

Findings

At 75% coverage, ORI initiated in week 1 averted a median of 129 cases (IQR 41–375; 70% of total). Impact declined exponentially with delay—by 15% per week and by half every 4.2 weeks—falling to 22 cases (9.8%) by week 9, comparable to historical response times. Early ORIs averted 0.69 cases (IQR 0.19–2.40) per 1,000 doses at US\$4,659 per case averted; week 9 responses achieved 0.11 cases per 1,000 doses at US\$13,039 per case. Outbreaks in the top quartile of size or attack rate yielded up to sixfold greater impact and sevenfold higher efficiency. When aggregated across outbreaks reflecting OCV stockpile use, week 9 ORIs still achieved 0.44 (IQR 0.40–0.47) cases averted per 1,000 doses, rising to 2.13 (IQR 2.08 - 2.15) when initiated in week 1 among the top 70% of outbreaks by attack rate.

Conclusion

Rapid OCV deployment and prioritization of large, sustained, high-attack-rate outbreaks are critical to maximize impact. Even with delays, stockpile-supported campaigns can substantially reduce cholera burden in Africa.

Key messages

1. We analyzed 1,192 cholera outbreaks across sub-Saharan Africa (2010–2020) to estimate the health and economic impact of outbreak response immunizations (ORIs) using single-dose oral cholera vaccine (OCV).
2. When ORI was implemented within one week of outbreak onset, 129 cases (IQR: 41 – 375; 70% [IQR:49.8 - 82.5]), cases were averted; however, impact declined sharply, halving each four weeks of delay.
3. Prioritizing large, prolonged, and high-attack-rate outbreaks yielded markedly impact, efficiency, and cost-effectiveness.
4. When simulations reflected historical OCV stockpile deployments, even 9-week-delayed ORIs still averted 0·44 (IQR 0·40–0·47) cases per 1,000 doses. These values increased to 2·13 (IQR 2·08–2·15) when campaigns were initiated in week 1 among the top 70% of outbreaks by attack rate.
5. Timely and targeted ORI deployment is essential to maximize the impact of limited global OCV supply and reduce cholera burden in Africa.

Research in context

Evidence before this study

We searched PubMed on March 15, 2024, with no language or date restrictions, using the search terms (("Cholera"[Mesh] OR "Vibrio cholerae"[Mesh])) AND "Cholera Vaccines"[Mesh] AND ((model*[Title/Abstract]) OR (math*[Title/Abstract])) AND

(vaccine*[Title/Abstract]), yielding 164 articles. Of these, 17 studies met the inclusion criteria, focusing on cholera transmission modeling to evaluate vaccine impact under various scenarios. Most studies simulated outbreaks in single, unique geographic setting¹⁻¹³, while only a few articles examined outbreaks across more than one geographic setting.¹⁴⁻¹⁷ These studies varied significantly in campaign type, including reactive mass campaigns, proactive campaigns, “hotspot” or area-targeted campaigns. Variability in outbreak size, duration, and modeling parameters—such as vaccine delivery and immunogenicity delays—were also common. For example, while some authors assumed protection immediately following vaccination^{9,13-15}, others considered a lag between vaccination and protection.^{1,6,8,10} Most prior models focused on large outbreaks (e.g., Zimbabwe with 98,591 cases or Haiti with 119,902 cases)^{8,15} and lacked small or short-duration outbreaks. These gaps highlight the need for models that reflect a broader spectrum of outbreak characteristics to enhance the generalizability of vaccine impact assessments.

Added value of this study

To our knowledge, this is the first study to model cholera vaccine impact across a large, diverse dataset of outbreaks from multiple temporal and geographic contexts in sub-Saharan Africa. By analyzing 1,406 unique outbreaks from 2010 to 2023—ranging in size, duration, attack rate, and population dynamics—our model offers a comprehensive and generalizable framework for evaluating oral cholera vaccine (OCV) impact in outbreak response immunizations (ORIs). Unlike previous studies focused on isolated large outbreaks, our approach captures the variability of outbreaks in their responses to vaccination timing and coverage. This broader modeling framework provides insights into cases and deaths averted and the cost-effectiveness of ORI campaigns across different outbreak characteristics and resource allocation strategies.

Implications of the available evidence

Our study offers critical insights for guiding global cholera vaccine implementation strategy. We demonstrate that ORI effectiveness declines by half approximately every four weeks

during the first 15 weeks of delay, underscoring the importance of timely campaign implementation. Despite these delays, ORIs remain highly effective for large, prolonged outbreaks, even when initiated as late as nine weeks post-outbreak onset. For example, with a 9-week delay, the median number of cases averted was 6 (<5%) across all outbreaks, whereas the largest 10% of outbreaks saw a median of 1,338 cases averted (>40%). When the vaccine impact is assessed across outbreaks collectively, overall percent reduction in cases reaches 27.7% [IQR: 25.1, 30.2] and 0.435 [IQR:0.401 - 0.477] cases are averted per 1,000 OCVs. These impacts can be further amplified if ORI campaigns are prioritized for large outbreaks. These findings highlight the need for targeted vaccination strategies that prioritize high-burden outbreaks, enabling more efficient resource allocation and maximizing the overall impact of limited OCV supplies.

Main text

Introduction

Cholera, one of the oldest infectious diseases affecting humans, has persisted as a major public health challenge since the 7th pandemic began near the Bay of Bengal in 1961.¹⁸ It remains a significant threat in settings with inadequate water, sanitation, and hygiene (WaSH), particularly in low- and middle-income countries (LMICs).¹⁹ Modelled estimates and World Health Organization (WHO) reports indicate that sub-Saharan Africa carries the majority of the global burden, marked by substantial spatial heterogeneity.^{20–22} Analysis of cholera incidence in the region from 2010 to 2016 identified nearly 1,000 outbreaks of varying scale across multiple administrative levels,²³ underscoring the complexity of controlling transmission.

Sustained access to adequate WaSH—along with case management and surveillance—remains the foundation of cholera control. However, the oral cholera vaccine (OCV) provides an essential complementary tool, particularly where long-term infrastructure improvements are difficult. To support rapid vaccine access, WHO established a global

OCV stockpile in 2013 with Gavi's support.²⁴ Between 2013 and 2024, more than 200 million doses have been deployed for both proactive and reactive campaigns.²⁵ In response to rising demand and constrained supply, the International Coordinating Group (ICG)—which manages the stockpile for WHO and Gavi—temporarily adopted a single-dose strategy for outbreak response in October 2022, replacing the standard two-dose schedule.²⁶

Given limited resources, optimizing vaccine use is critical. Although targeting high-risk areas preemptively based on environmental covariates or historical incidence²⁶, or sensitive diagnostics,¹⁷ can improve benefits, such approaches are challenged by fluctuating incidence,²⁷ waning immunity,²⁸ and population mobility.^{9,29} In contrast, outbreak response immunizations (ORIs)—vaccinating populations where outbreaks occur—reduce uncertainties tied to preemptive targeting. In practice, 77% of over 200 million doses deployed by 2024 were used for ORIs rather than prevention.²⁵

Despite this widespread use, evidence on ORI impact remains fragmented. Most previous analyses examined a few large outbreaks,^{1,3,5,6,10,15,23} which fail to represent the diverse epidemiological profiles of cholera outbreaks,^{23,30} or overlook key operational and immunological factors such as delayed campaign initiation^{25,31} or time to develop vaccine-derived immunity.^{9,13–15} A comprehensive modeling framework that captures the full epidemiological diversity of outbreaks, real-world immunological delays, and operational constraints is therefore essential to inform cholera control strategies.

We addressed this evidence gap by evaluating the health and economic impact of single-dose OCV ORIs across 1,192 cholera outbreaks in sub-Saharan Africa from 2010 to 2020. Specifically, we asked: (1) how effective are single-dose OCV ORIs when implemented under varying delays and coverage levels during individual outbreaks? and (2) how effective are ORIs when deployed across multiple heterogeneous outbreaks, as in global stockpile operations? To answer these questions, we combined outbreak-level impact estimates with

aggregate analyses simulating vaccine deployment under stockpile constraints across a range of implementation scenarios.

Methods

Outbreak data

We compiled weekly suspected cholera incidence data from the Cholera Taxonomy Data,³² which includes 1,664 outbreaks reported across various administrative levels in 30 countries in sub-Saharan Africa from January 2010 to July 2023.^{23,33} Outbreaks were defined by location-specific average weekly incidence thresholds, spanning from the first to the last reported suspected case, with weeks without data assumed to have zero cases.³⁰ To ensure consistency with real-world outbreak response conditions, some outbreaks were excluded. We excluded outbreaks reported from administrative units with populations outside the range of historical OCV deployments—specifically, those corresponding to fewer than 16,500 or more than 5 million doses.³⁴ Also, when outbreaks overlapped across multiple administrative levels, smaller nested outbreaks entirely contained within a larger one were excluded. If a smaller administrative unit reported an equal or greater number of cases, the smaller outbreak was retained and the larger one excluded. In addition, outbreaks occurring within three years after an OCV campaign were removed to avoid vaccine-influenced estimates, assuming that these outbreaks may have already been influenced by OCVs and therefore do not serve as good controls for counterfactual analyses. Finally, outbreaks reported after 1 January 2020 were excluded due to unavailable OCV campaign data. After excluding these 472 outbreaks, 1,192 outbreaks were retained for modeling. (Figure 1A).

The outcomes were reported as cases, deaths, and disability-adjusted life years (DALYs) averted, which were also converted to vaccination campaign efficiency metrics (e.g., cases, deaths, and DALYs averted per 1,000 OCVs) and cost-effectiveness metrics (e.g., cost per case, death, or DALY averted)

Modeling the vaccine impact

We simulated weekly case counts under counterfactual vaccination scenarios—identical in all respects except for the implementation of ORIs. The impact of ORIs was based on the number of cases averted through vaccination (**Figure 1B**, top panel). Because ORIs can only occur if outbreaks are still ongoing, we assumed no campaign would be initiated if two or more weeks had passed since the outbreak ended, consistent with the outbreak definition³⁵ and validated in sensitivity analyses. ORIs beginning before outbreak resolution may still produce no impact due to the time required for implementation and vaccine-derived immunity to develop (**Figure 1B**, second panel). The impact of ORIs tends to be larger for a larger outbreak (third panel of **Figure 1B**) and the overall impact across multiple heterogeneous outbreaks tends to be larger than the simple average, as larger outbreaks contribute disproportionately to the overall impact (fourth panel in **Figure 1B**).

The model captures both direct and indirect effects of OCV³⁶ (Supplementary Information **S1**) and incorporates age-specific vaccine efficacy (< 5 and 5+ yo), delays in ORI implementation, and the delays in the vaccine-derived immunity (**Table 1**). The proportion of cases averted, $P_{i,\pi}$, at a given vaccine coverage π for each age group—children under five ($i = 1$) and individuals aged five or older ($i = 2$)—was computed as:

$$P_{i,\pi} = 1 - [\pi(1 - \psi_i)(1 - \eta_\pi) + (1 - \pi)(1 - \eta_\pi)],$$

where ψ_i represents direct vaccine efficacy for age group i and η_π denotes the indirect vaccine efficacy at coverage π . Thus, unvaccinated individuals, comprising $1 - \pi$ of the population, still benefit from indirect protection $1 - \eta_\pi$. We estimated η_π from reduction in incidence rates among unvaccinated individuals observed in a randomized trial conducted in Matlab, Bangladesh using Bayesian beta regression.^{11,37} (Supplementary Information **S2** and **Figure S1**). The cumulative number of cases, $C_{i,\pi,\tau}$, for age group i , under a vaccination campaign initiated at time τ with coverage π was calculated as:

$$C_{i,\pi,\tau} = \sum_{t=1}^T I_{t,i} [1 - \mathbf{1}(t \geq \tau + \delta) P_{i,\pi}],$$

where $I_{t,i}$ is the incidence at time t for age group i and $\mathbf{1}(x)$ is an indicator function that equals 1 if x is true and 0 otherwise. The term δ captures delays for campaign implementation and the development of vaccine-derived immunity, and T denotes the outbreak's end date. Overall, we simulated ORIs in each of the 1,192 outbreaks, exploring various vaccination coverage levels, campaign timing, and trigger criteria (outbreak onset-triggered or case-triggered) (**Table 2**). We sampled 200 parameter sets via a Sobol low-discrepancy sequence (**Table 3** and Supplementary Information **S3**), yielding varying estimates of ORI impact. Results are reported as medians with interquartile range (IQR).

In addition to evaluating the impact of ORIs at the level of individual outbreaks, we also estimated their overall impact by aggregating the effects across multiple outbreaks under a constrained OCV supply. Specifically, we modeled the deployment of approximately 100,000 to 200 million doses of OCVs administered in reactive campaigns globally, loosely based on OCV doses deployed through OCV stockpile.³⁴ Because this supply was insufficient to cover all outbreaks in the dataset, we simulated vaccine allocation across all outbreaks or subsets of outbreaks selected by such features of outbreaks as attack rate, duration, size, and case fatality ratio. Specifically, we randomly sampled outbreaks (without replacement) and assigned each enough vaccine to cover its target population, continuing this process until the number of doses deployed was matched. This entire process was iterated 200 times. For each iteration, we calculated the overall impact by aggregating the outcome variables across sampled outbreaks. These results were then summarized to estimate the overall impact.

Our modeling approach focuses on the immediate impact of ORI during ongoing outbreaks and does not account for potential additional benefits, such as the prevention of future outbreaks in the same population. This assumption reflects the limited duration of population-level protection expected from single-dose campaigns, given the waning of

immunity conferred by single-dose regimens and population mobility.³⁸ Moreover, we modeled each population independently, without considering potential cross-population effects from outbreaks or vaccination campaigns. All code and datasets used in the analysis are publicly available in a GitHub repository.³⁹

Table 1. Model parameters.

Description	Values	Source
<i>Parameters varied</i>		
Direct vaccine efficacy by one dose of OCV for the < 5 yo and 5+ yo, respectively.	30% [95% CI: 15%, 42%] (< 5 yo) and 64% [95% CI: 58%, 70%] (5+ yo)	28
Campaign duration	5 [range: 3-14] days	31
Indirect vaccine efficacy	77.9% [95% CrI: 54.5 – 91] and 95% [95% CrI: 58.4 – 99.6] at 60% and 90% vaccine coverages, respectively. (Supplementary Information S3 and Figure S1)	
Delay in vaccine-derived immunity	10.5 [range: 3, 21] days (Supplementary Information S4)	40–48
<i>Parameters fixed</i>		
Vaccine coverage	60%, 75%, and 90%	
Proportion of moderate and severe cholera	0.78, 0.22	49
Disability weight for moderate and severe diarrhea	0.188, 0.247	50
Vaccine cost of procurement	US\$ 1.65 per dose	51
Vaccine delivery cost	US\$ 2.32 per dose	52
Vaccine shipping cost	US\$ 0.06 per dose	53
Cost to patient for cholera hospitalization and outpatient services	US\$ 109.32 and US\$ 16.06 per patient	54

Cost to public system for cholera hospitalization and outpatient services	US\$ 184.97 and US\$ 7.99 per patient	54
Number of workdays lost to patients and caregivers	6.5 and 3.3 days (Supplementary Information S5)	55–58
Mean age of infection	25 years (Supplementary Information S6)	59–61
Duration of illness	2 days	62
Life expectancy at a given age	Varies by year and country; 42.7 [IQR: 42.2 – 43.2] year of life expectancy for 25-year-olds during 2010-2020.	63
Age distribution	Varies by year and country; 16.4% [IQR: 16.0, 16.9] of the population is under five during 2010 - 2020.	63
Gross Domestic Product (GDP)	Varies by year and country; 1 665 [IQR: 1 621 – 1 706] US\$ during 2010-2020	64
Population in workforce	Varies by year and country; around 68.1% [IQR: 67.9 - 68.4] in during 2010-2020	65

Campaign triggers and assumptions

ORIs were triggered either at the outbreak onset (outbreak onset-triggered) or when the cumulative number of reported symptomatic cases reached certain thresholds (case-triggered) (**Table 2**). The outbreak onset-triggered approach highlights how campaign delays influence the impact of ORI, offering insights into the interplay between the timing and the impact of ORI. The case-triggered approach adds additional **potentially realistic delays** in which the exact start date may be uncertain (e.g., because of poor surveillance) and an ORI is triggered only after some threshold number of cases is reached. Under the case-triggered approach, the same cumulative-case threshold could trigger ORIs earlier in more rapidly growing outbreaks and later in more slowly growing outbreaks.

Vaccine coverage levels were varied from 60%, 75%, to 90% of the affected population—an administrative unit where the outbreak was reported. The vaccine coverages were chosen based on trends observed in global OCV campaigns (2013-2018)⁶⁶ and a recent campaign in Ethiopia.⁶⁷ We selected 75% coverage as a representative value for a single-dose regimen, reflecting estimated coverage levels typically observed during OCV responses in humanitarian crises or outbreaks (Supplementary Information **S7**). The delay in protective immunity post-vaccination was chosen reflecting the time required for seroconversion as a proxy^{40–48}, which occur between 3 to 28 days (most often 7 to 14 days) following the first dose in a two-dose regimen^{40,42–45} (Supplementary Information **S4**).

Table 2. Model settings for ORI (single-dose OCV campaign) settings

OCV campaign feature	Model settings that were tested
Campaign coverage	Targeted the entire population of the affected administrative area. Coverage ranged from 60%, 75%, to 90%.
Campaign triggers	(i) Outbreak onset–triggered: Initiated in 1-15 weeks of outbreaks, in 1-week increments; implementation set at the midpoint of the specified week. (ii) Case-triggered: Initiated once cumulative cases reached a predefined threshold (1, 5, 10, 30, 50, 70, 90, 140, 190, 240, 290), followed by an additional 1–15 week implementation delay.
Vaccination rounds	One round of vaccination per campaign
Outbreak data	Analyses were performed across all outbreaks at the individual outbreak level and in aggregated fashion in subsets that were selected by attack rate, duration, size, and case fatality ratio.

Cost-effectiveness analysis

We evaluated cost-effectiveness by calculating the cost per case, death, and DALY averted. Following a literature review (**Table 1** and Supplementary Material **S8**), we selected recent, high-quality cost estimates representative of our modeling data. To be comprehensive, we

adopted a societal perspective that includes direct costs (e.g., vaccine procurement and healthcare utilization) and indirect costs (e.g., productivity losses). This perspective provides a holistic view of ORI interventions. All costs were adjusted to 2023 US\$ equivalents by inflating the costs from the original reporting year based on annual consumer price index changes for sub-Saharan Africa through 2023.^{68–70}

Results

Outbreak data

We identified 1,192 cholera outbreaks across sub-Saharan Africa between 2010 and 2020. At the continental level, cholera incidence displayed marked temporal and spatial heterogeneity. Monthly time series (**Figure 2A**) show that most outbreaks occurred in Eastern, Western, and Central Africa. Outbreaks during the first half of the period (2010–2015) were concentrated in Western Africa, while those in the latter half (2016–2021) were more prevalent in Eastern Africa. In contrast, Central Africa experienced outbreaks more consistently across the study period. Spatial heterogeneity was also evident at finer geographic scales, such as at the country level (**Figure 2B**). Outbreaks typically peaked around week 5 (mean = 6 weeks and median = 4 weeks), with approximately 88% reaching their peak before week 10 (**Figure 2C**). Substantial variation was observed in outbreak characteristics, including size duration, and attack rate (**Figures 2D–E**). Outbreak durations ranged from 5 to 93 weeks, sizes from 3 to 14,522 suspected cases, and the attack rates from 0.005 to 112 rates per 10,000 populations. Of the 152 outbreaks reporting at least one laboratory-confirmed case, duration ranged from 5 to 56 weeks, outbreak sizes from 10 to 7,726 suspected cases, and attack rates from 0.005 to 19 rates per 10,000 population. Excluded outbreaks exhibited similar spatiotemporal patterns, although their sizes tended to be larger. These consistencies between included and excluded outbreaks suggest that the dataset is broadly representative (**Figure S2** in Supplementary Material).

Impact of ORIs in individual outbreaks

Impact of ORIs across campaign timing and coverage levels

In the outbreak-onset-triggered scenario, an ORI launched without delay (week 1) at 75% coverage averted 70% [IQR: 49.8-82.5] of all outbreak cases—corresponding to 129 cases [IQR: 41-375] averted when counted over the entire outbreak (that is, cases occurring before and after the trigger relative to a no-ORI counterfactual; **Figure 3A**). Higher vaccine coverage consistently yielded greater benefit: at week 1, the median numbers of cases averted were 118, 130, and 136 for coverage levels of 60%, 75%, and 90%, respectively. By week 9, these medians fell to 26, 28, and 30, and the median percent cases averted dropped below 13% across all coverage levels. Substantial variation remained across outbreaks for each delay, reflecting heterogeneity in outbreak dynamics and ORI parameters. Exponential decay model captured the decline in impact across delay ($R^2=0.942$ [IQR: 0.882 - 0.975]) (**Figure 3B**). Among outbreaks lasting >9 weeks, the instantaneous reduction rate was 0.166 per week [IQR: 0.079 - 0.293], implying a week-to-week decline of 15.3% [IQR: 7.6 - 25.4] and a halving time of 4.2 weeks [IQR: 2.4 - 8.8]. Decay was faster in subsets enriched for shorter outbreaks (i.e., 0.193 among outbreaks lasting >5 weeks). Delay also causes some outbreaks to be missed entirely because vaccination would not occur before the outbreak ends (**Figure 3C**). From week 5 onward, an additional 5.3% of outbreaks were missed per week of delay; by week 9, ~26.6% of potential ORIs would be missed.

Impact of vaccine varied by epidemiological characteristics of outbreaks and different metrics were influenced to different extents. Prioritizing larger outbreaks produced the largest gains in the number of cases averted per ORI (**Figure 3D**; **Figure S3** in the **Supplement**) whereas prioritizing high attack rate outbreaks produced the largest gains in vaccine efficiency—measured as cases or DALYs averted per 1,000 OCV doses (or people as a single-dose regimen) (**Figures 3E–F**). Targeting top 10% of outbreaks by outbreak size increased the median to 691 case compared to 21.8 cases when randomly targeted with

ORI initiated at week 9. This value increases to 1,725 cases if ORI could be initiated in the first week. Prioritizing top 10 % of outbreak regarding attack rate increased the vaccination efficiency approximately to 2.5 (or 0.07 DALYs) and 9.6 cases (1.3 DALYs) averted per 1,000 doses with ORIs initiated in week 9 and 1, respectively. Prioritizing by size or duration also improved efficiency, although gains were smaller than targeting by attack rate (**Figure S3**). In contrast, the percent cases averted (which is not normalized by doses) was most sensitive to prioritizing by duration. When ORIs were initiated in week 1, the median percent cases averted were approximately 83.8%, 82%, and 80.1% for outbreaks in the top decile of duration, size, and attack rate, respectively. These values fell to 55.2%, 27.2%, and 18.3%, respectively, by week 9.

Table 3 summarizes the impact of ORI under the outbreak onset-triggered scenario with 75% vaccine coverage (results for other coverage levels and delays are provided **Table S1**). Consistent with the pattern of cases averted shown in **Figure 3**, the impact of ORI diminished across other health and economic metrics, including DALYs averted, and efficiency measured as cases and DALYs averted per 1,000 OCV doses, while the cost per case, or DALY averted increased with longer delays. The cost per case averted was approximately 4,659 [IQR: 983 - 17,915] for ORIs that were randomly targeted, initiated in the first week, this cost rose markedly to \$15,744 [IQR: 2,911 - 75,908]. As expected in increased OCV efficiency under high AR prioritized targeting, the cost per case averted was 414 [IQR: -17 - 871] and 1,733 [IQR: 689 - 5,229] for ORI initiated at week 1 and 9, respectively. The cost per DALY averted were \$24,763 [IQR: 4,059 - 4,676], much higher than approximate GDP per capita in sub-Saharan Africa ($\approx \$1,636^{64}$) in 2010 - 2020.

Table 3 also shows that ORI impact, OCV efficiency, and cost effectiveness were consistently greater in the top quartile of outbreaks by attack rate, duration, or size. Percent case averted in the top quartile of outbreaks by duration reached 82% [IQR: 72.9 – 88] and 24.6% [IQR: 12.9 - 44.1] when ORI was launched in week 1 and week 13, respectively. OCV efficiency and cost effectiveness were more pronounced in the top quartile of outbreaks by

attack rate. Focusing on outbreaks of top quartile of attack rate can achieve 2,967 [IQR: -37 - 45,357] and 8,533 [IQR: 1,292 - 1,186,542] for ORIs implemented in week 1 and 9, respectively. The median values were comparable to approximately 1.8 and 5.2 times the GDP per capita values. For the case-triggered scenario, the impact of ORI declined with longer delays after the trigger and higher thresholds (**Table S2**). Efficiency also decreased with higher thresholds and longer delays. Accordingly, the median cost per case, death, and DALY averted rose with higher thresholds and longer delays, remaining below conventional cost-effectiveness benchmarks.

Impact of ORIs aggregated across multiple outbreaks

Impact of ORIs across campaign timing and coverage levels

Assuming 100,000 to 200 million OCV doses, the number of outbreaks eligible for ORI ranged from 1 to >530 when targeted at random. Because outbreaks differ in population affected, the number of outbreaks targeted decreased when prioritizing larger outbreaks (approximately 480 outbreaks) and increased when prioritizing high-attack-rate (AR) outbreaks ($\approx$ 800 outbreaks), which are usually smaller (**Figure 4A**). Total cases averted increased with both the number of doses deployed and the prioritization criteria (**Figure 4B**). With a 9-week delay in ORI implementation, the median number of cases averted reached 68,000 [IQR: 65,900–75,800] under random targeting, compared with 87,000 [78,100–95,700], 97,000 [89,900–103,800], and 156,000 [154,100–158,300] when prioritizing outbreaks with larger size, longer duration, and higher AR, respectively. When the delay was shortened to 1 week, the corresponding numbers increased to 194,000 [184,400–202,500], 244,000 [236,200–253,700], 259,000 [243,000–276,200], and 420,000 [418,200–423,600] for the same four strategies.

Efficiency and cost effectiveness

Efficiency—measured as cases averted per 1,000 OCV doses—showed large variability at low vaccine supply (100,000 and 200,000 doses) but converged at higher coverage (**Figure 4C**). As the number of doses increased, targeting high-AR outbreaks consistently produced the highest efficiency, exceeding 2 cases averted per 1,000 doses with a 1-week delay and about 1 case per 1,000 doses with a 9-week delay, while other strategies remained lower. The pattern for DALYs averted per 1,000 doses mirrored that of cases averted but with smaller magnitude—approximately 0.6 for high-AR targeting at a 1-week delay and 0.3–0.4 for other strategies (**Figure 4D**). Economic efficiency also improved markedly with both shorter delays and better targeting. The cost per case averted declined from US\$8,285 [7,776–9,191] under random targeting to US\$4,299 [4,231–4,421] when high-AR outbreaks were prioritized with a 9-week delay, and further to US\$3,346 [3,106–3,505] and US\$1,331 [1,312–1,357] under a 1-week delay for random and high-AR targeting, respectively (**Figure 4E**). Similarly, cost per DALY averted decreased from US\$31,248 [28,347–36,212] to US\$16,621 [16,279–16,812] with a 9-week delay, and from US\$10,997 [10,253–11,610] to US\$4,695 [4,602–4,753] with a 1-week delay (**Figure 4F**). Together, these results indicate that targeting outbreaks with high attack rates offers the largest gains in both health and economic outcomes, and that timeliness of ORI implementation remains a critical determinant of OCV impact.

When the impact and efficiency of ORI campaigns with 75% vaccine coverage was evaluated across aggregated sets of outbreaks, median effectiveness and efficiency increased with the number of outbreaks included in the sample (**Figure S4**). As such substantial impact was observed when the outbreaks were aggregated to reflect the volume of OCVs deployed through the global stockpile (**Figure 4**; more details in **Table S3**). Over time, the aggregated number of cases averted by these campaigns increased, reaching 4,028 [IQR: 3,679 - 4,376] by 2020. The percentage cases averted also rose steadily to 32.3% [IQR: 29.1 - 35.2] in 2020, assuming a 9-week delay in ORI implementation with 75%

coverage. These values increase further if ORIs are prioritized to longer outbreaks. If outbreaks are targeted to top 25% of outbreaks by duration, case averted increased to 7,805 [IQR: 7,126 - 8,438], corresponding to 41.3% [IQR: 37.6 - 44.8] under the same conditions of 9-week delay in ORI implementation with 70% coverage. These estimates across aggregated outbreaks contrast with the lower median impact observed when vaccination scenarios were evaluated at the level of individual outbreaks—e.g., 12.3 [IQR: 1.7 - 34.2] under a comparable implementation delay (as previously illustrated in **Table 2** and **Figure 3**).

Efficiency and cost effectiveness

Cases averted per 1,000 OCVs (**Figure 5D**) tended to increase while the cost per case averted (**Figure 5E**) and the cost per DALY averted (**Figure 5F**) tended to decrease as the years went by and the number of OCV doses used increased. Also, variability in these metrics decreased as the number of deployed doses increased. By 2020, the estimated case averted per 1,000 OCVs were 0.409 [IQR: 0.374 - 0.451], which could increase to 0.736 [IQR: 0.663 - 0.813] if top quartile of outbreaks by duration are prioritized. These median values are substantially higher than 0.151 [IQR: 0.013 - 0.892] per 1,000 OCVs when the estimate was evaluated at the individual outbreak level. The cost per case averted were US\$ 9,035 [IQR: 8,142 - 9,945], which could be reduced to 4,498 [IQR: 4,017 - 5,054] if top quartile of outbreaks by duration are prioritized. These values again are much lower than US\$ 13,045 [IQR: 2,546 - 59,533] when evaluated for individual outbreaks (as previously shown in **Table 3**). Similarly, the cost per DALY averted was US\$ 27,307 [IQR: 23,727 - 31,423], which could decrease to US\$ 10,695 [IQR: 9,807 - 11,761] when ORIs were prioritized to the top quartile of outbreak by duration. The median values are again much lower than US\$ 1,744,390 [IQR: 11,650 - 40,068,121] when evaluated for individual outbreaks (as previously shown in **Table 3**).

Discussion

Our study demonstrated the substantial potential impact of OCVs to reduce the burden of cholera outbreaks, based on modeling of 1,192 outbreaks in sub-Saharan Africa from 2010 to 2020. We found that the timing of ORI is critical: the median percentage reduction in cases declined from 70.0% to 7.8% when ORI was delayed from week 1 to week 9. This corresponds to approximately 15.3% reduction in case averted per week of delay in vaccination, with the benefit halving approximately every 4.2 weeks. Vaccine efficiency showed a similar decline, dropping from roughly 0.7 to 0.1 cases averted per 1,000 vaccinated persons over the same delay interval.

Our findings also highlight that impact of ORI depends not only on campaign parameters but also on outbreak characteristics. Delays in campaign initiation can be offset by targeting outbreaks that are likely to be longer, larger, or have high AR in the absence of intervention. For example, median case reduction (approximately 22.8%) achieved by an ORI with 5-week delay across all outbreaks is comparable to that of a 13-week delay in outbreaks in the top quartile by duration (23.3%). Similarly, OCV efficiency was 0.2 for an ORI with 5-week delay across all outbreaks whereas it was 0.5 in the top quartile by AR. This underscores the value of strategic prioritization: directing limited OCV supplies to outbreaks with a higher likelihood of being large, prolonged, or high AR can significantly amplify the overall effect and efficiency of ORIs. While this approach assumes an idealized ability to predict outbreak size or duration at the time of response, such prediction may be partially feasible using known risk factors—such as environmental indicators, socioeconomic conditions (e.g., poverty, literacy, healthcare access), and WASH infrastructure.^{19,71–74} These proxies can help guide the allocation of limited OCV supplies toward outbreaks with the greatest potential for impact and efficiency. Prioritizing high-risk geographic areas has similarly been shown to improve the effectiveness and efficiency of OCV campaigns in sub-Saharan Africa.⁷⁵ In addition, highly targeted interventions such as case-area targeted interventions have demonstrated enhanced OCV efficiency.¹ As predictive models and surveillance systems continue to improve, the capacity to forecast

outbreak characteristics in real time will support more effective and targeted vaccination strategies.

Our studies also illustrate that the prioritizing outbreaks that can produce high impact and efficiency is also as critical as reducing the delay in ORI implementation, which manifested as large uncertainty intervals in plots presented throughout. As an illustration, percent case averted can vary from zero to close to 100 % regardless of delays in ORIs implementation. Similarly, OCV efficiency could vary from 0 to 27.5 and 144 at ORI implemented at week 9 and 1, respectively.

Our study also illustrate the median impact at the individual level does not translate to the overall impact if multiple outbreaks of different characteristics are targeted over time. Targeting outbreaks repeatedly eventually ran across large, long-lasting, and high AR outbreaks, which tend to produce higher impact and efficiency, contribute disproportionately to the overall impact. Assuming 200 million OCV doses—comparable to the total stockpile deployed globally between 2013 and 2024—our model estimated around median 0.4 cases (compared to approximately 0.1 at the individual outbreak level). With prioritization in favor of attack rates, our model estimated up to 420,000 cases averted when vaccination was initiated at week 1, equivalent to approximately 2.1 cases averted per 1,000 doses administered.

While estimates from previous studies are quite diverse, our estimates for the impact, efficiency, and cost effectiveness appear comparable to existing estimates and we believe that our estimates provide more realistic versions. In contrast to our extensive dataset, most previous studies assessing OCV impact during or prior to cholera outbreaks have focused on single outbreaks or a small number of large-scale events.^{1-4,6-8,10,16,76} For example, a preemptive vaccination campaign in Matlab, Bangladesh with 50% coverage was associated with a 93% reduction in cholera cases [95% CI: 82% - 99%].¹¹ Another modeling study of reactive evaluating single-dose OCV campaigns in Zimbabwe, Haiti, and

Guinea found case reductions of 71.6%, 65.3%, and 61.9%, when campaigns were initiated on day 122, day 76, and day 61, respectively.¹⁵ Similarly, a single-dose OCV campaign in N'Djamena, Chad reduced cases by 81% (or 43% or 8%) when initiated on day 50 (or day 130 or day 180)¹ Additional modeling of multiple strategies in N'Djamena, Maela (Thailand), and Haiti found that a single-dose regimen targeting individuals over five years old often produced greatest reductions, lowering total and symptomatic infections by up to 68%, 90%, and 36%, respectively¹⁴. Together, these studies demonstrate the large potential of OCV campaigns, but focusing on large outbreaks may bias estimates upward.

OCV efficiency (the number of cases averted per 1,000 OCV doses) was also higher in some previous studies focused on large outbreaks in Matlab, Bangladesh, Haiti, or a few African cities [ref]. The increased estimates can be attributed to, among many other things that involve other modeling assumptions, to again that prior studies focused on single or few relative larger, high AR outbreaks that tend to produce high OCV efficiency. They also employ two dose schedule with higher vaccine efficacy. They also employ pre-emptive vaccination strategies or with reactive vaccination while not incorporating the delay in the development of vaccine-derived immunity. The study by Lee et al. provides more comparable setting than other previous studies by incorporating heterogeneities in cholera transmission (while not focusing on outbreaks) in sub-Saharan Africa. The study assumes preventive campaigns under endemic transmission using two-dose schedule. It found out that with some selective geographical targeting provides estimates varied from 2.6 (95% PI 2.4–2.9) to 12 (95% PI 10.3–14.4) per 1,000 fully vaccinated persons. Our estimates of 2.1 appear comparable considering that we assumed single-dose regimen and therefore dividing the estimates by Lee et al. by 2 would be fairer comparison. Also, our estimates shows cumulative effect of 200,000 million doses which include around 800 outbreaks (although OCV efficiency of this magnitude is already observed with 1 million doses) and not all outbreaks has AR as high as the incidence rate that Lee et al. study used. In other words, it is straightforward to find a condition that can produce OCV efficiency estimates as high as the highest estimates by Lee et al. by focusing on high AR outbreaks. Similarly, cost

effectiveness measures, measured as cost per DALY averted, is comparable to existing estimates. The study by Lee et al, again shows that \$1,843 (95% CI 1,032–2,382) to \$21,213 (95% CI 11,953–our estimate show that around \$4,695 with ORI implemented at week 1

Also, defining outbreaks based on the reported may be a larger

And most studies do not account for the delays in vaccination response as we incorporated in our

.

Despite its strengths, our study has several limitations. First, we assumed uniform vaccine coverage within administrative units, which may underestimate the benefits of more spatially focused strategies such as case-area targeted interventions^{1,77–79} or geographic targeting⁷⁵ if lower-level, high-risk administrative units are targeted than our model assumed tion of targeting all people of the “outbreaks-affected” administrative unit. population. Conversely, vaccine wastage⁸⁰ could lead to overestimates of ORI efficiency and cost effectiveness. Second, our parameterization of indirect vaccine effectiveness was derived from studies conducted in Bangladesh,^{11,37} and may not generalize fully to sub-Saharan African settings. Nonetheless, the consistency between our static model outputs and those from dynamic models (Supplementary Material **S9; Figures S5-S7**) suggests that our parameterization captures indirect effects reasonably well. This static approach, grounded in empirical observations, may offer a practical alternative for modeling ORI impact in settings with limited data. Third, by modeling individual outbreak independently (or aggregating them), we assumed that the vaccine impact was confined to the specific outbreak in which vaccines were deployed. In reality, ORIs may also confer preemptive benefits by preventing future outbreaks in the same or neighboring areas, given two-dose killed whole-cell OCVs provide overall protection for up to 4-5 years.^{81–84} Due to a lack of geographic details for each outbreak and epidemiologic links between the outbreaks, we could not evaluate the downstream or spatial spillover effects. Fourth, our reliance on

suspected rather than confirmed cases may overestimate the vaccine impact given that roughly half may not be true cholera.⁸⁵ On the other hand, underreporting of cholera in sub-Saharan Africa may offset this discrepancy⁸⁶. Fifth, we assumed that ORIs did not occur if the delay in implementation exceeded the duration of the outbreak—that is, if the outbreak had ended two weeks before the ORI was initiated. However, in real-world settings, it is often difficult to determine in real time whether an outbreak has concluded and ORIs may be conducted anyways. Under this alternative scenario, the estimated impact of ORIs declined more sharply, as expected from the fact that zero-impact ORIs are included in the calculation: the median percentage reduction in cases with a 9-week delay dropped from 12.3% in the main scenario to 2.7% in the alternative (Supplementary Material **Table SX**)

Sixth, our estimates are contingent on the outbreak dataset used, which is based on a specific outbreak definition that is not universally accepted and may vary by context. Applying alternative outbreak criteria could yield different datasets and, consequently, different estimates of vaccine impact. The main analysis excluded outbreaks that were either too small or too large, based on the population sizes commonly observed in historical OCV campaigns. While this filtering was appropriate for retrospective evaluation, such thresholds may not be applicable to future scenarios. To explore a broader range of outbreak settings, we relaxed these criteria and included outbreaks occurring in populations between 5,000 and 15,000,000, instead of the original range of 16,500 to 5,000,000. Under this more inclusive scenario, the median and interquartile ranges remained largely unchanged (**Table S4**). Finally (or Seventh), the overall impact, efficiency, and cost effectiveness derived from our simulation of OCV deployment from stockpile requires careful interpretation. Our simulation ignores many potentially important realistic details to produce accurate estimation of the overall impact of OCV deployment from global stockpile. That is, our simulation does not account for, for example, critical characteristics of actual outbreaks targeted, vaccine coverage levels, detailed distribution of delays in ORI implementation. Our simulation of overall simulation was intended to illustrate the disproportionate contribution of larger and longer outbreaks to the overall

impact and prioritization, which eventually will increase the median impact we can expect as we continue to vaccinate the outbreak-affected population even with delays.

Conclusion

This study provides important insights into the effectiveness of outbreak response immunizations (ORIs) across a broad range of cholera outbreak scenarios in sub-Saharan Africa. Our findings underscore the critical importance of timely intervention and strategic outbreak prioritization to maximize vaccine impact. Looking ahead, advances in predictive modeling, the integration of real-time surveillance data, and the development of more flexible and targeted vaccine deployment strategies will be essential to contribute meaningfully to global cholera control and elimination efforts.

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Author contributions

J-HK and ECL conceptualized the study. J-HK led the study design and analysis. J-HK and MD performed the statistical analysis and drafted the initial manuscript. ECL and QZ collected and prepared the data for analysis. ECL, ASA, and JHK provided critical input on methodology. MD conducted the literature review. All authors participated in the

interpretation of the findings, critically reviewed the manuscript for intellectual content, and approved the final version for submission.

References

1. Finger F, Bertuzzo E, Luquero FJ, Naibei N, Touré B, Allan M, et al. The potential impact of case-area targeted interventions in response to cholera outbreaks: A modeling study. Von Seidlein L, editor. *PLOS Med*. 2018 Feb 27;15(2):e1002509.
2. Mukandavire Z, Manangazira P, Nyabadza F, Cuadros DF, Musuka G, Morris JG. Stemming cholera tides in Zimbabwe through mass vaccination. *Int J Infect Dis*. 2020 Jul;96:222–7.
3. Dimitrov DT, Troeger C, Halloran ME, Longini IM, Chao DL. Comparative Effectiveness of Different Strategies of Oral Cholera Vaccination in Bangladesh: A Modeling Study. Baker S, editor. *PLoS Negl Trop Dis*. 2014 Dec 4;8(12):e3343.
4. Khan AI, Levin A, Chao DL, DeRoeck D, Dimitrov DT, Khan JAM, et al. The impact and cost-effectiveness of controlling cholera through the use of oral cholera vaccines in urban Bangladesh: A disease modeling and economic analysis. Maheu-Giroux M, editor. *PLoS Negl Trop Dis*. 2018 Oct 9;12(10):e0006652.
5. Lee EC, Chao DL, Lemaitre JC, Matrajt L, Pasetto D, Perez-Saez J, et al. Achieving coordinated national immunity and cholera elimination in Haiti through vaccination: a modelling study. *Lancet Glob Health*. 2020 Aug;8(8):e1081–9.
6. Chao DL, Halloran ME, Longini IM. Vaccination strategies for epidemic cholera in Haiti with implications for the developing world. *Proc Natl Acad Sci*. 2011 Apr 26;108(17):7081–5.
7. Kirpich A, Weppelmann TA, Yang Y, Morris JG, Longini IM. Controlling cholera in the Ouest Department of Haiti using oral vaccines. Coudeville L, editor. *PLoS Negl Trop Dis*. 2017 Apr 14;11(4):e0005482.
8. Havumaki J, Meza R, Phares CR, Date K, Eisenberg MC. Comparing alternative cholera vaccination strategies in Maela refugee camp: using a transmission model in public health practice. *BMC Infect Dis*. 2019 Dec;19(1):1075.
9. Peak CM, Reilly AL, Azman AS, Buckee CO. Prolonging herd immunity to cholera via vaccination: Accounting for human mobility and waning vaccine effects. Marks F, editor. *PLoS Negl Trop Dis*. 2018 Feb 28;12(2):e0006257.
10. Azman AS, Luquero FJ, Rodrigues A, Palma PP, Grais RF, Banga CN, et al. Urban Cholera Transmission Hotspots and Their Implications for Reactive Vaccination: Evidence from

Bissau City, Guinea Bissau. Vinetz JM, editor. PLoS Negl Trop Dis. 2012 Nov 8;6(11):e1901.

11. Longini IM, Nizam A, Ali M, Yunus M, Shenvi N, Clemens JD. Controlling Endemic Cholera with Oral Vaccines. Von Seidlein L, editor. PLoS Med. 2007 Nov 27;4(11):e336.
12. Smalley HK, Keskinocak P, Swann J, Hinman A. Optimized oral cholera vaccine distribution strategies to minimize disease incidence: A mixed integer programming model and analysis of a Bangladesh scenario. Vaccine. 2015 Nov;33(46):6218–23.
13. Azman AS, Lessler J. Reactive vaccination in the presence of disease hotspots. Proc R Soc B Biol Sci. 2015 Jan 7;282(1798):20141341.
14. Leung T, Eaton J, Matrajt L. Optimizing one-dose and two-dose cholera vaccine allocation in outbreak settings: A modeling study. Fleckenstein JM, editor. PLoS Negl Trop Dis. 2022 Apr 20;16(4):e0010358.
15. Azman AS, Luquero FJ, Ciglenecki I, Grais RF, Sack DA, Lessler J. The Impact of a One-Dose versus Two-Dose Oral Cholera Vaccine Regimen in Outbreak Settings: A Modeling Study. Kretzschmar MEE, editor. PLOS Med. 2015 Aug 25;12(8):e1001867.
16. Kim JH, Mogasale V, Burgess C, Wierzbza TF. Impact of oral cholera vaccines in cholera-endemic countries: A mathematical modeling study. Vaccine. 2016 Apr;34(18):2113–20.
17. Xu H, Zou K, Dent J, Wiens KE, Malembaka EB, Bwire G, et al. Enhanced cholera surveillance to improve vaccination campaign efficiency. Nat Med [Internet]. 2024 Mar 5 [cited 2024 Apr 16]; Available from: <https://www.nature.com/articles/s41591-024-02852-8>
18. Kaper JB, Morris JG, Levine MM. Cholera. Clin Microbiol Rev. 1995 Jan;8(1):48–86.
19. Sikder M, Deshpande A, Hegde ST, Malembaka EB, Gallandat K, Reiner RC, et al. Water, Sanitation, and Cholera in Sub-Saharan Africa. Environ Sci Technol. 2023 Jul 18;57(28):10185–92.
20. Ali M, Nelson AR, Lopez AL, Sack DA. Updated Global Burden of Cholera in Endemic Countries. PLoS Negl Trop Dis. 2015 Jun 4;9(6):e0003832.
21. Clemens JD, Nair GB, Ahmed T, Qadri F, Holmgren J. Cholera. Lancet Lond Engl. 2017 Sep 23;390(10101):1539–49.
22. Lessler J, Moore SM, Luquero FJ, McKay HS, Grais R, Henkens M, et al. Mapping the burden of cholera in sub-Saharan Africa and implications for control: an analysis of data across geographical scales. Lancet Lond Engl. 2018 May 12;391(10133):1908–15.

23. Zheng Q, Luquero FJ, Ciglenecki I, Wamala JF, Abubakar A, Welo P, et al. Cholera outbreaks in sub-Saharan Africa during 2010-2019: a descriptive analysis. *Int J Infect Dis.* 2022 Sep 1;122:215–21.
24. Yen C, Hyde TB, Costa AJ, Fernandez K, Tam JS, Hugonnet S, et al. The development of global vaccine stockpiles. *Lancet Infect Dis.* 2015 Mar 1;15(3):340–7.
25. Global Task Force on Cholera Control. GTFCC OCV Dashboard [Internet]. Available from: <https://apps.epicentre-msf.org/public/app/gtfcc>
26. Lee EC, Azman AS, Kaminsky J, Moore SM, McKay HS, Lessler J. The projected impact of geographic targeting of oral cholera vaccination in sub-Saharan Africa: A modeling study. *PLoS Med.* 2019 Dec;16(12):e1003003.
27. Perez-Saez J, Lessler J, Lee EC, Luquero FJ, Malembaka EB, Finger F, et al. The seasonality of cholera in sub-Saharan Africa: a statistical modelling study. *Lancet Glob Health.* 2022 Jun 1;10(6):e831–9.
28. Bi Q, Ferreras E, Pezzoli L, Legros D, Ivers LC, Date K, et al. Protection against cholera from killed whole-cell oral cholera vaccines: a systematic review and meta-analysis. *Lancet Infect Dis.* 2017 Oct 1;17(10):1080–8.
29. Meredith HR, Giles JR, Perez-Saez J, Mande T, Rinaldo A, Mutembo S, et al. Characterizing human mobility patterns in rural settings of sub-Saharan Africa. *eLife.* 2021 Sep 17;10:e68441.
30. Zheng Q, Lee EC. Cholera outbreak time series in Africa between 2010 and 2023. In preparation;
31. Pezzoli L, Oral Cholera Vaccine Working Group of the Global Task Force on Cholera Control. Global oral cholera vaccine use, 2013-2018. *Vaccine.* 2020 Feb 29;38 Suppl 1(Suppl 1):A132–40.
32. Johns Hopkins Cholera Dynamics Team. CholeraTaxonomy [Internet]. [cited 2025 Jan 2]. Available from: <https://cholera-taxonomy.middle-distance.com/>
33. Zheng Q, DePencier R, Demby MN, Azman AS, Lee EC. Cholera surveillance time series in Africa between 2010 and 2023. *VeriXiv.* 2025;2(116).
34. GTFCC OCV Dashboard [Internet]. [cited 2024 Nov 7]. Available from: <https://apps.epicentre-msf.org/public/app/gtfcc>
35. Cholera surveillance time series in Africa between 2010 ... | VeriXiv [Internet]. [cited 2025 Jul 9]. Available from: <https://verixiv.org/articles/2-116>

36. Halloran ME, Struchiner CJ. Study designs for dependent happenings. *Epidemiol Camb Mass*. 1991 Sep;2(5):331–8.
37. Ali M, Emch M, Von Seidlein L, Yunus M, Sack DA, Rao M, et al. Herd immunity conferred by killed oral cholera vaccines in Bangladesh: a reanalysis. *The Lancet*. 2005 Jul;366(9479):44–9.
38. Peak CM, Reilly AL, Azman AS, Buckee CO. Prolonging herd immunity to cholera via vaccination: Accounting for human mobility and waning vaccine effects. *PLoS Negl Trop Dis*. 2018 Feb 28;12(2):e0006257.
39. Kim JH. cholera-outbreak-vaccine-impact-modeling [Internet]. 2024. Available from: <https://github.com/kimfinale/cholera-outbreak-vaccine-impact-modeling>
40. Ng Ombe H, Simuyandi M, Mwaba J, Luchen CC, Alabi P, Chilyabanyama ON, et al. Immunogenicity and waning immunity from the oral cholera vaccine (Shanchol™) in adults residing in Lukanga Swamps of Zambia. *PloS One*. 2022;17(1):e0262239.
41. Kanungo S, Paisley A, Lopez AL, Bhattacharya M, Manna B, Kim DR, et al. Immune responses following one and two doses of the reformulated, bivalent, killed, whole-cell, oral cholera vaccine among adults and children in Kolkata, India: a randomized, placebo-controlled trial. *Vaccine*. 2009 Nov 16;27(49):6887–93.
42. Chowdhury F, Bhuiyan TR, Akter A, Bhuiyan MS, Khan AI, Hossain M, et al. Immunogenicity of a killed bivalent whole cell oral cholera vaccine in forcibly displaced Myanmar nationals in Cox’s Bazar, Bangladesh. *PLoS Negl Trop Dis*. 2020 Mar;14(3):e0007989.
43. Raghava Mohan V, Raj S, Dhingra MS, Aloysia D’Cor N, Singh AP, Saluja T, et al. Safety and immunogenicity of a killed bivalent (O1 and O139) whole-cell oral cholera vaccine in adults and children in Vellore, South India. Borrow R, editor. *PLOS ONE*. 2019 Jun 18;14(6):e0218033.
44. Charles RC, Hilaire IJ, Mayo-Smith LM, Teng JE, Jerome JG, Franke MF, et al. Immunogenicity of a killed bivalent (O1 and O139) whole cell oral cholera vaccine, Shanchol, in Haiti. *PLoS Negl Trop Dis*. 2014 May;8(5):e2828.
45. Chac D, Bhuiyan TR, Saha A, Alam MM, Salma U, Jahan N, et al. Gut Microbiota and Development of *Vibrio cholerae*-Specific Long-Term Memory B Cells in Adults after Whole-Cell Killed Oral Cholera Vaccine. *Infect Immun*. 2021 Aug 16;89(9):e0021721.
46. McCarty JM, Cassie D, Bedell L, Lock MD, Bennett S. Long-Term Immunogenicity of Live Oral Cholera Vaccine CVD 103-HgR in Adolescents Aged 12-17 Years in the United States. *Am J Trop Med Hyg*. 2021 Apr 5;104(5):1758–60.

47. McCarty JM, Lock MD, Hunt KM, Simon JK, Gurwith M. Safety and immunogenicity of single-dose live oral cholera vaccine strain CVD 103-HgR in healthy adults age 18–45. *Vaccine*. 2018 Feb;36(6):833–40.
48. Chowdhury F, Ali Syed K, Akter A, Rahman Bhuiyan T, Tauheed I, Khaton F, et al. A phase I/II study to evaluate safety, tolerability and immunogenicity of Hillchol®, an inactivated single Hikojima strain based oral cholera vaccine, in a sequentially age descending population in Bangladesh. *Vaccine*. 2021 Jul 22;39(32):4450–7.
49. Mbewe N, Imamura T, Sivile S, Sinkala A, Zulu P, Chanda C, et al. Clinical Characteristics of 6578 Adult Patients With Cholera Admitted to Community and Referral Cholera Treatment Centers in Lusaka, Zambia, October 2023 to April 2024. *Open Forum Infect Dis*. 2025 Jun 1;12(6):ofaf277.
50. GBD 2017 Disease and Injury Incidence and Prevalence Collaborators. Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet Lond Engl*. 2018 Nov 10;392(10159):1789–858.
51. UNICEF. Oral cholera vaccine (OCV) price data [Internet]. 2024 Oct. Available from: <https://www.unicef.org/supply/media/22781/file/OCV-vaccines-prices-29102024.pdf>
52. Sarker AR, Khan AI, Islam MT, Chowdhury F, Khanam F, Kang S, et al. Cost of oral cholera vaccine delivery in a mass immunization program for children in urban Bangladesh. *Vaccine X*. 2022 Dec;12:100247.
53. UNICEF. Emergency stockpile availability - oral cholera vaccine [Internet]. 2024 Nov. Available from: <https://www.unicef.org/supply/media/22091/file/EmergencyStockpileAvailabilityReportOCV11112024.pdf>
54. Mogasale V, Ngogoyo SM, Mogasale VV. Model-based estimation of the economic burden of cholera in Africa. *BMJ Open*. 2021 Mar;11(3):e044615.
55. Ilboudo PG, Huang XX, Ngwira B, Mwanyungwe A, Mogasale V, Mengel MA, et al. Cost-of-illness of cholera to households and health facilities in rural Malawi. *PloS One*. 2017;12(9):e0185041.
56. Awalime DK, Davies-Teye BBK, Vanotoo LA, Owoo NS, Nketiah-Amponsah E. Economic evaluation of 2014 cholera outbreak in Ghana: a household cost analysis. *Health Econ Rev*. 2017 Dec 4;7(1):45.
57. Poulos C, Riewpaiboon A, Stewart JF, Clemens J, Guh S, Agtini M, et al. Costs of Illness Due to Endemic Cholera. *Epidemiol Infect*. 2012 Mar;140(3):10.1017/S0950268811000513.

58. Schaetti C, Weiss MG, Ali SM, Chaignat CL, Khatib AM, Reyburn R, et al. Costs of illness due to cholera, costs of immunization and cost-effectiveness of an oral cholera mass vaccination campaign in Zanzibar. *PLoS Negl Trop Dis*. 2012;6(10):e1844.
59. Ohene-Adjei K, Kenu E, Bando DA, Addo PNO, Noora CL, Nortey P, et al. Epidemiological link of a major cholera outbreak in Greater Accra region of Ghana, 2014. *BMC Public Health*. 2017 Oct 11;17(1):801.
60. Azman AS, Parker LA, Rumunu J, Tadesse F, Grandesso F, Deng LL, et al. Effectiveness of one dose of oral cholera vaccine in response to an outbreak: a case-cohort study. *Lancet Glob Health*. 2016 Nov 1;4(11):e856–63.
61. Luquero FJ, Grout L, Ciglenecki I, Sakoba K, Traore B, Heile M, et al. Use of *Vibrio cholerae* Vaccine in an Outbreak in Guinea. *N Engl J Med*. 2014 May 29;370(22):2111–20.
62. Weil AA, Khan AI, Chowdhury F, Larocque RC, Faruque ASG, Ryan ET, et al. Clinical outcomes in household contacts of patients with cholera in Bangladesh. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2009 Nov 15;49(10):1473–9.
63. United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects 2024, Online Edition [Internet]. 2024. Available from: <https://population.un.org/wpp/Download/Standard/Mortality/>
64. World Bank Group. GDP per capita (current US\$) [Internet]. Available from: <https://data.worldbank.org/indicator/NY.GDP.PCAP.CD>
65. World Bank Group. Labor force participation rate, total (% of total population ages 15–64) (modeled ILO estimate) [Internet]. Available from: <https://data.worldbank.org/indicator/SL.TLF.ACTI.ZS?end=2023&start=1960>
66. Pezzoli L. Global oral cholera vaccine use, 2013–2018. *Vaccine*. 2020 Feb;38:A132–40.
67. Edosa M, Jeon Y, Gedefaw A, Hailu D, Mesfin Getachew E, Mogeni OD, et al. Comprehensive Review on the Use of Oral Cholera Vaccine (OCV) in Ethiopia: 2019 to 2023. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2024 Jul 12;79(Suppl 1):S20–32.
68. Mogasale V, Ramani E, Wee H, Kim JH. Oral Cholera Vaccination Delivery Cost in Low- and Middle-Income Countries: An Analysis Based on Systematic Review. Ryan ET, editor. *PLoS Negl Trop Dis*. 2016 Dec 8;10(12):e0005124.
69. Turner HC, Lauer JA, Tran BX, Teerawattananon Y, Jit M. Adjusting for Inflation and Currency Changes Within Health Economic Studies. *Value Health*. 2019 Sep;22(9):1026–32.

70. Bureau of Labor Statistics. Math calculations to better utilize CPI data [Internet]. Bureau of Labor Statistics; Available from: <https://www.bls.gov/cpi/factsheets/cpi-math-calculations.pdf>
71. Giroto CD, Behzadian K, Musah A, Chen AS, Djordjević S, Nichols G, et al. Analysis of environmental factors influencing endemic cholera risks in sub-Saharan Africa. *Sci Total Environ*. 2024 May 20;926:171896.
72. Koepke AA, Longini IM, Halloran ME, Wakefield J, Minin VN. PREDICTIVE MODELING OF CHOLERA OUTBREAKS IN BANGLADESH. *Ann Appl Stat*. 2016 Jun;10(2):575–95.
73. Armando CJ, Rocklöv J, Sidat M, Tozan Y, Mavume AF, Bunker A, et al. Spatial-temporal analysis of climate and socioeconomic conditions on cholera incidence in Mozambique from 2000 to 2018: an ecological longitudinal retrospective study. *BMJ Open*. 2024 Aug;14(8):e082503.
74. Ali M, Sen Gupta S, Arora N, Khasnobis P, Venkatesh S, Sur D, et al. Identification of burden hotspots and risk factors for cholera in India: An observational study. Moise IK, editor. *PLOS ONE*. 2017 Aug 24;12(8):e0183100.
75. Lee EC, Azman AS, Kaminsky J, Moore SM, McKay HS, Lessler J. The projected impact of geographic targeting of oral cholera vaccination in sub-Saharan Africa: A modeling study. Kretzschmar MEE, editor. *PLOS Med*. 2019 Dec 11;16(12):e1003003.
76. Delport D, Muellenmeister AM, MacKechnie G, Vaccher S, Mengistu T, Hogan D, et al. Estimating the historical impact of outbreak response immunization programs across 210 outbreaks in LMICs [Internet]. *medRxiv*; 2024 [cited 2025 Feb 21]. p. 2024.06.02.24308241. Available from: <https://www.medrxiv.org/content/10.1101/2024.06.02.24308241v1>
77. Sikder M, Altare C, Doocy S, Trowbridge D, Kaur G, Kaushal N, et al. Case-area targeted preventive interventions to interrupt cholera transmission: Current implementation practices and lessons learned. *PLoS Negl Trop Dis*. 2021 Dec;15(12):e0010042.
78. O'Keeffe J, Salem-Bango L, Desjardins MR, Lantagne D, Altare C, Kaur G, et al. Case-area targeted interventions during a large-scale cholera epidemic: A prospective cohort study in Northeast Nigeria. *PLoS Med*. 2024 May;21(5):e1004404.
79. Kaur G, Salem-Bango L, Nery ALMDS, Solomon EC, Ihemezue E, Kelly C, et al. Implementation considerations in case-area targeted interventions to prevent cholera transmission in Northeast Nigeria: A qualitative analysis. *PLoS Negl Trop Dis*. 2023 Apr;17(4):e0011298.
80. Lessler J, Metcalf CJE, Grais RF, Luquero FJ, Cummings DAT, Grenfell BT. Measuring the Performance of Vaccination Programs Using Cross-Sectional Surveys: A Likelihood Framework and Retrospective Analysis. *PLOS Med*. 2011 25;8(10):e1001110.

81. Ali M, Qadri F, Kim DR, Islam T, Im J, Ahmmed F, et al. Unmasking herd protection by an oral cholera vaccine in a cluster-randomized trial. *Int J Epidemiol*. 2019 Aug 1;48(4):1252–61.
82. Khatib AM, Ali M, Seidlein L von, Kim DR, Hashim R, Reyburn R, et al. Effectiveness of an oral cholera vaccine in Zanzibar: findings from a mass vaccination campaign and observational cohort study. *Lancet Infect Dis*. 2012 Nov 1;12(11):837–44.
83. Ali M, Sur D, You YA, Kanungo S, Sah B, Manna B, et al. Herd Protection by a Bivalent Killed Whole-Cell Oral Cholera Vaccine in the Slums of Kolkata, India. *Clin Infect Dis*. 2013 Apr 15;56(8):1123–31.
84. Ali M, Emch M, Seidlein L von, Yunus M, Sack DA, Rao M, et al. Herd immunity conferred by killed oral cholera vaccines in Bangladesh: a reanalysis. *The Lancet*. 2005 Jul 2;366(9479):44–9.
85. Wiens KE, Xu H, Zou K, Mwaba J, Lessler J, Malembaka EB, et al. Estimating the proportion of clinically suspected cholera cases that are true *Vibrio cholerae* infections: A systematic review and meta-analysis. *PLoS Med*. 2023 Sep;20(9):e1004286.
86. Wiens KE, Miller MH, Costello DJ, Solomon AP, Hilbert SM, Shipper AG, et al. Care-seeking for diarrheal illness: A systematic review and meta-analysis. *PLOS Glob Public Health*. 2025 Apr 22;5(4):e0004483.